Examiner only

Section A

Answer **all** questions in the spaces provided.

Refer to the separate Resource Folder to answer question **1**.

1. (a) (i) The Red Admiral butterfly competes with greenfly for food. The scientific name for one type of greenfly is *Acyrthosiphon pisum.*

Complete the table below to show the scientific classification of greenfly. [3]

Kingdom:	....
Phylum:	....
Class:	Insecta
Order:	Hemiptera
Family:	Aphididae
Genus:	....
Species:	....
Scientific name:	*Acyrthosiphon pisum*

(ii) Another butterfly is found in Africa. Its scientific name is *Vanessa abyssinica.*

I. State **one** advantage of using scientific names. [1]

....

....

II. State **one** piece of evidence, from its scientific name, that shows it is related to the Red Admiral. [1]

....

....

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(b) The variation in population of adult Red Admirals is shown by the graph in **Figure 1**.

(i) I. State the name of the seasonal movement of animals as seen in Red Admirals. [1]

....

II. State **one** reason why this seasonal movement occurs. [1]

....

....

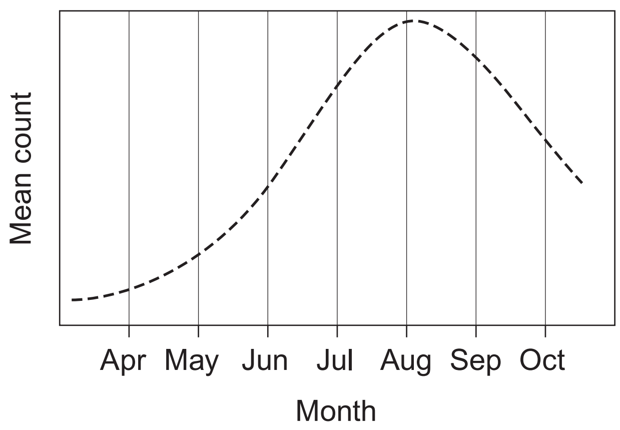
(ii) Use the information in the graph in **Figure 1** to compare the variation of the Red Admiral population in 2019 with the mean change since 1976. [2]

....

....

....

(iii) Red Admirals are a food source for frogs. **Add a line** to the graph below to show how the population of frogs changed between April and October in 2019. [2]



(iv) Use your line to explain the relationship between numbers of predators (frogs) and the numbers of their prey (Red Admirals). [3]

....

....

....

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(c) (i) Use the information in **Table 2** to construct the part of the food web that depends on berries. [3]

Berries

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(ii) Explain why the frog can be found at more than one trophic level. [2]

....

....

....

(d) Refer to **Table 2** to construct a pyramid of biomass for a food chain of **five** trophic levels starting with berries and ending with buzzards. [3]

(e) Use the information in **Figure 2** to calculate the efficiency of energy transfer between the primary and secondary consumers. [3]

% efficiency = ...

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